

# Mini Project 2: House Sales Price Prediction (King County Dataset)

## Project Overview

This project focuses on analyzing and predicting **house prices** using a real-world dataset from **King County, USA**. The dataset contains detailed information about property features such as size, location, condition, and sale price. The objective is to clean the data, perform exploratory data analysis (EDA), visualize key patterns, and build a **basic regression model** to predict house prices.

## Project Objectives

1. Load and explore the dataset
2. Preprocess and clean the data
3. Perform Exploratory Data Analysis (EDA)
4. Create visualizations for insights
5. Build a simple Machine Learning model
6. Summarize insights and model performance

## Insights & Summary

### Key Insights

1. House prices increase significantly with **sqft\_living**.
2. Properties with higher **grade** and better **condition** command higher prices.
3. Waterfront houses are priced much higher on average.
4. Older houses generally have lower prices unless renovated.
5. Location (latitude & longitude) plays an important role in pricing.

### Model Performance Summary

- Linear Regression explains a significant portion of price variance.
- $R^2$  score indicates reasonable prediction capability.
- Model can be improved using advanced algorithms.

## Conclusion

This project demonstrated how machine learning can be applied to real estate price prediction. Through proper data cleaning, EDA, visualization, and regression modeling, we were able to understand key factors affecting house prices and build a baseline predictive model.